MUHAMMAD AZBIN FAHMI BIN HAIRUL FAIZAL

Full-Stack Web Developer (Web Platforms · APIs · Real-Time Systems)
Sungai Petani, Kedah
azbinfahmi@gmail.com | +60 1156516348
Portfolio: azbinfahmi.com



SUMMARY

Full-stack web developer with 2+ years of professional experience building production web platforms, backend APIs, and real-time systems. Strong in Node.js, TypeScript, PostgreSQL, and WebSocket, with hands-on experience in IoT-driven applications, and browser-based 3D visualization. Proven ability to design end-to-end systems from data ingestion to frontend delivery.

WORK EXPERIENCE

June 2024 - Ongoing

Full-Stack Web Developer (Remote) | GoSpatial Tech Pte Ltd

- Designed and developed **production-grade web platforms** using Node.js, TypeScript, REST APIs, WebSocket, and PostgreSQL.
- Built **backend services** to ingest, process, validate, and store high-volume real-time data from IoT devices and gateways.
- Developed **real-time dashboards and web interfaces** for monitoring system status, device telemetry, and event-driven alerts.
- Designed and maintained **WebSocket-based real-time communication** for live updates, streaming data, and alert propagation.
- Implemented **custom API layers** to integrate third-party systems with internal platforms, enabling automation, analytics, and data visualization.
- Customized and extended the **ThingsBoard platform**, including custom widgets, dashboards, rule-engine logic, and backend integrations.
- Integrated **AI-powered services** (YOLOv8, Deepstack) into web systems for real-time detection and automated event triggering.
- Designed and built a **web-based data processing platform** enabling users to upload large datasets, trigger backend workflows, and visualize results directly in the browser.
- Implemented **background processing pipelines** and data transformation workflows for large-file handling and 3D web visualization.
- Collaborated remotely with cross-functional teams to deliver features end-to-end from requirements to production deployment.

Feb 2023 – June 2024

Drafter (Drafting & Internal Web Tools) | Echo Broadband Sdn. Bhd.

- Worked as a drafter on **COAX/Fiber and Gas pipeline network projects**, contributing to the design and validation of large-scale telecommunications and utility infrastructure.
- Identified **manual, time-consuming validation workflows** in fiber network design and proactively designed a **custom internal web application** to automate error detection.
- Developed a **fiber network validation web tool** to detect missing connections, inconsistent topology, and mismatches between planned and as-built data.
- Replaced manual splitter-by-splitter checks with automated validation logic, **significantly reducing review time and human error**.
- Applied the same automation approach to **GAS Malaysia projects**, improving coordinate validation accuracy and internal review workflows.
- Conducted quality control reviews and on-site assessments to validate data accuracy and implementation.

- **Received an innovation award** for independently developing internal web tools that improved productivity, accuracy, and project turnaround time. (View Award Certificate)

TECHNICAL SKILLS

Languages
JavaScript, TypeScript

Frontend
HTML, CSS, JavaScript, AngularJS, Web-based 3D Visualization

Backend & APIs
Node.js, REST APIs, WebSocket, PostgreSQL, MQTT, HTTP

Systems & Platforms
ThingsBoard, Real-Time Systems, Data Pipelines, Background Processing

DevOps & Tools
Docker, Git

LANGUAGES

Bahasa Melayu
English

EDUCATION

Bachelor of Science (Geoinformatics)
Universiti Teknologi Malaysia
CGPA: 3.42 / 4.00 | Dean's List (2 Semesters)
